

Optimal Design of Water Tank Size for Power System Flexibility and Water Quality

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Abstract—The increasing penetration of renewable energy, such as wind power, has brought great challenges to the power system operation due to its uncertainty. Flexibility, which measures the ability of power system to deal with the uncertainties, is critical for power system to adapt to the new era of renewable energy. The rising electrical demand of the water distribution system (WDS) creates opportunities for power system to leverage the flexibility provided by WDS. This paper investigates the water-energy relationship between these two systems and makes full use of the water pumps and tanks in the WDS to enhance the power system flexibility. Current WDS designs did not consider well the role WDS plays in the power system. This hinders the WDS from being fully used by power system to provide flexibility. An optimization model is proposed to determine the optimal tank size of WDS, which may provide the maximum available flexibility of power system. The effect of tank size on water quality is also investigated to ensure that the supplied water quality is not compromised. Moreover, a Benders-based heuristic algorithm is proposed to find the optimization solution more efficiently and to protect the data of each energy system. Results of case study highlight the merit of the proposed optimization design and the advantage of using WDS to provide flexibility for power system.

Index Terms—Wind power, power system, water distribution system, flexibility, optimization.

NOMENCLATURE

Most of the symbols and notations used throughout this paper are defined below. Others are defined following their first appearances, as needed.

A. Sets

$\mathbb{X}/\mathbb{U}/\mathbb{Y}$ Set of variables involved in the outer/middle/inner objective function of two-stage robust optimization:
 $\mathbb{X} = \{I_{s,i,t}, U_{s,i,t}, D_{s,i,t}\}$,

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$\mathbb{U} = \{R_{s,w,t}^{WN}, R_{s,n,t}^{EL}\}$, and $\mathbb{Y} = \{P_{s,w,t}^{wc}, P_{s,n,t}^{ec}, P_{s,i,t}^g\}$
 $\mathbb{Z}$ Set of variables involved in the objective function of tank size optimization
 Ω^T/Ω^W Set of conventional generating units/wind farms
 Ω^L Set of blocks in the piecewise function of the conventional generating unit
 Λ^E/Ψ^E Set of electrical buses/lines
 $\Omega_n^T/\Omega_n^W/\Omega_n^{WDS}$ Set of units/wind farms/WDSs that connected to bus n
 Λ_n^E Set of electrical buses connected to bus n
 $\Delta/\Delta_P/\Delta_V$ Set of water pipes/pumps/valves
 $\varpi/\varpi_T/\Omega^{WDS}$ Set of water nodes/tanks/WDS
 Δ^+ Set of water pipes where the flow direction is from the starting node to the end node

B. Parameters

N_{sc} Number of seasonal scenarios
 ω_s Scenario occurrence probability
 N_g Number of conventional units
 T Number of time steps
 c_i, c_i^u, c_i^d Commitment/start-up/shut-down cost of unit i
 N_W/N_{EB} Number of wind farms/electrical loads
 λ/μ Penalty factor for wind power/electrical demand curtailment
 L Number of piecewise linear cost function segments
 $K_{i,l}$ Slope of block l of the piecewise function of unit i
 TU_i/TD_i Minimum up/down times of unit i
 $Ramp_i$ Ramp rate of unit i
 P_i^{gmin}/P_i^{gmax} Minimum/maximum output of unit i
 $P_{s,w,t}^{pre}$ Output of wind farm w at time t in scenario s
 $d_{s,n,t}$ Demand of electrical load n at time t in scenario s
 $B_{n,m}$ Susceptance of electrical line $n-m$
 $P_{s,w,t}^{WN,down}/R_{s,w,t}^{WN,up}$ Lower/upper bound of the output fluctuation range of wind farm w at time t in scenario s

$R_{s,n,t}^{EL,down} / R_{s,n,t}^{EL,up}$	Lower/upper bound of the demand fluctuation range of electrical load n at time t in scenario s	C. Variables	
$PF_{n,m}$	Capacity limit of electrical transmission line $n-m$	$I_{s,i,t} / U_{s,i,t} / D_{s,i,t}$	Binary variable for commitment/startup/shutdown status of unit i at time t in scenario s (equals to 1 when the unit is on/started up/shut down, and 0 otherwise)
N_{WDS}	Number of WDS	$P_{s,w,t}^{wc} / P_{s,n,t}^{ec}$	Amount of wind power curtailment in the wind farm w /electrical demand curtailment at node n at time t in scenario s
N_h^{WT}	Number of tanks in WDS h	$P_{s,i,t,l}^g$	Power produced in block l of the piecewise linear function of the unit i at time t in scenario s
ϖ_h^n	Unit construction cost of tank n in WDS h	$R_{s,w,t}^{WN} / R_{s,n,t}^{EL}$	Uncertain proportion of output/demand in wind farm w /electrical load n at time t to the values of scenario s
r	Interest rate	$\theta_{s,n,t} / \theta_{s,ref,t}$	Phase angle of electrical node n /reference node at time t in scenario s
IY	Project life of the water tank in years	$WV_{s,h,t}^n$	Size of tank n in WDS h
$\lambda price_t$	Electricity price at time t	$P_{s,h,t}^{WDS}$	Electricity consumption of WDS h at time t in scenario s with the optimized tank size
ξ	Penalty factor	$h_{s,J_m,t}^t / h_{s,J_n,t}^t$	Head pressure of water node J_m / J_n at time t in scenario s
$P_{s,h,t}^{WDSO}$	Electricity used by WDS h at time t in scenario s without optimizing the tank size	$q_{s,mn,t}^t / q_{s,P_m,t}^t$	Flow in pipe $m-n$ /pump m at time t in scenario s
$P_{s,t}^{Flex}$	Flexibility requirement of power system at time t in scenario s	$\hat{h}_{s,P_m,t}^t$	Pressure head gain of pump m at time t in scenario s
c_{mn}	Loss coefficient of pipe $m-n$	$z_{s,mn,t}^t$	Binary variable which equals to 1 if the valve $m-n$ works at time t in scenario s and 0 otherwise
$q_{mn}^{min} / q_{mn}^{max}$	Minimum/maximum volume of pipe $m-n$	$R_{s,mn,t}^t$	Head loss along the pipe $m-n$ equipped with a valve at time t in scenario s
$a_{P_m}, b_{P_m}, c_{P_m}$	Coefficients of the head-flow equation of pump m	$q_{s,J_m,t}^{in} / q_{s,J_m,t}^{out}$	Water that flows into/out of tank m at time t in scenario s
$q_{P_m}^{min} / q_{P_m}^{max}$	Minimum/maximum value of the flow when pump m on	$W_{s,J_m,t}$	Volume of the stored water in tank m at time t in scenario s
M	A very large positive constant	$W_{J_m}^{wind}$	Excess volume for providing flexibility in tank m
$qv_{mn}^{min} / qv_{mn}^{max}$	Minimum/maximum pipe flow when valve $m-n$ on	$P_{s,i,t}^{WPP}$	Electricity consumption of pump i at time t in scenario s
dw_{s,J_k}^t	Demand of water node J_k at time t in scenario s	$C_{mn}(s, x, t)$	Concentration of the residual chlorine at position x of pipe $m-n$ at time t in scenario s
$h_{J_m}^{min}$	Minimum head of water node J_m	$v_{mn}(s, t)$	Flow velocity in pipe $m-n$ at time t in scenario s
$A_{J_m}^{tank}$	Surface area of tank m	$r[C_{mn}(s, x, t)]$	Reaction rate at position x of pipe $m-n$ at time t in scenario s
μ_W	Water loss rate of tank	$C_{J_m}(s, t)$	Nodal residual chlorine concentration of node J_m at time t in scenario s
$W_{J_m}^{min}$	Required minimum water storage of tank m	$r[C_{J_m}(s, t)]$	Reaction rate of tank m at time t in scenario s
$W_{J_m}^{guide}$	Required constructed size of tank m to meet guidelines	$P_{s,w,t}^{wca} / P_{s,n,t}^{eca}$	Amount of wind power curtailment in the wind farm w /electrical demand
$WV_{J_m}^{max}$	Maximum construction volume of tank m obeying technical limitations		
$\kappa_1, \kappa_2, \kappa_3$	Weighting coefficients		
N_h^{WP}	Number of pumps in WDS h		
S_{mn}	Cross-sectional area of pipe $m-n$		
k, k_b	Coefficient of pipe/tank reaction rate equation		
L_{nm}	Length of pipe $n-m$		
$Q_{s,J_m,ex}^t$	External flow into node J_m at time t in scenario s		
$C_{J_m,ex}(s, t)$	Concentration of the external flow which comes into node J_m at time t in scenario s		
Δt	Time step		
$A_{n,w}$	Equal 1 when wind farm w is connected to electrical bus n , and 0 otherwise		
$B_{n,i}$	Equal 1 when unit i is connected to electrical bus n , and 0 otherwise		

$\theta_{s,n,t}^{AJ}/\theta_{s,m,t}^{AJ}$

curtailment at node n at time t in scenario s with the optimized tank size
Adjusted phase angle of node n/m at time t in scenario s

 $\theta_{s,ref,t}^{AJ}$

Phase angle of the reference bus at time t in scenario s

 $\vartheta_{s,i,t}^{(1)}/\vartheta_{s,i,t}^{(2)}$

Non-negative KKT multipliers corresponding to the left side/right side of constraint (8)

 $\vartheta_{s,i,t}^{(3)}/\vartheta_{s,i,t}^{(4)}$

Non-negative KKT multipliers corresponding to the left side/right side of constraint (9)

 $\vartheta_{s,i,t,l}^{(5)}$

Non-negative KKT multiplier corresponding to the constraint (10)

 $\vartheta_{s,w,t}^{(6)}/\vartheta_{s,w,t}^{(7)}$

Non-negative KKT multipliers corresponding to the constraint (12)/(13)

 $\vartheta_{s,n,t}^{(8)}/\vartheta_{s,n,t}^{(9)}$

Non-negative KKT multipliers corresponding to the constraint (14)/(15)

 $\tau_{s,m,t}^{(1)}$

KKT multiplier corresponding to the constraint (11)

 $z_{s,i,t}^{(1)}, z_{s,i,t}^{(2)}, z_{s,i,t}^{(3)}$

Binary variables for handling bilinear constraints

 $z_{s,i,t}^{(4)}, z_{s,i,t,l}^{(5)}, z_{s,w,t}^{(6)}$

Binary variables for handling bilinear constraints

 $z_{s,w,t}^{(7)}, z_{s,n,t}^{(8)}, z_{s,n,t}^{(9)}$

Binary variables for handling bilinear constraints

I. INTRODUCTION

RENEWABLE energy is estimated to meet 80% of global electricity demand growth during the next decade and is expected to overtake the coal by 2025 as the primary means of producing electricity [1]. Renewable energy such as wind power, however, is uncertain and highly volatile compared with the power generated by conventional thermal generating units. With the increasing penetration of wind power, great challenges on the imbalance of supply and demand have put forward high requirements for the flexibility of power system.

Flexibility is referred to as the capability of power system to cope with the uncertainty that arises from variable renewable generation [2]. Power system requires adequate flexibility to cope with the changing wind power. If not, events of wind power curtailment or electrical demand curtailment will occur and the power system may face huge economic losses. Inadequate flexibility is the main cause of the curtailment [3]. Thus, power system flexibility has become a global priority, as stated by International Energy Agency in their 2018 report [4]. It is in urgent need of finding effective means to ensure the flexible operation of power system.

WDS has been used to regulate the electricity consumption curve and promote the accommodation of wind power by coordinating the operation of water pumps and tanks [5]. WDS accounts for 5% of the annual total electricity consumption in China, and more than 15% in California, USA [6]. Electrical demand associated with WDS is projected to increase. There are a few pilot programs utilizing WDS to provide flexibility for

power system [7]–[11]. WDS can provide frequency regulation services, reduce the peak load, and balance the wind power integration. In California, a water utility, which owns a WDS and a wastewater treatment plant, can provide 5.3 MW interruptible load capacity for up to 2 hours, resulting in more than \$400000 annual revenues for the utility [9]. Thus, it is of great benefit to fully make use of the WDS to improve the flexibility of power system and to enhance the wind power accommodation.

Optimal operational strategies of integrating WDS into the power system have been studied [12]–[16]. As an essential component of the WDS, water tank plays an important role in the optimal schedule of electricity consumption. The WDS can pump water into the tank for storage when the wind is abundant and release the stored water to meet the demand. Water tank can be regarded as an auxiliary storage device for the power system. The optimal tank design scheme may provide the maximum available flexibility of power system.

The water tank design in WDS has been widely studied. For example, the pros and cons of various tank design models were analyzed [17]. The network reduction approach, hydraulic simulation, and mathematical optimization were combined to support the planning process for tanks [18]. The size and location of the water tank could be optimized by incorporating the Epanet solver into the decision process [19]. The optimization model of WDS is formulated as a mixed-integer quadratically-constrained programming (MIQCP) problem, which can be solved efficiently by heuristic algorithms [20]. Multi-objective optimization methods have been applied to the tank design considering factors such as cost, greenhouse gas emissions and reliability [21]–[24]. The above tank design schemes mainly consider the characteristics of WDS, and the effect of WDS connected to power system has not been taken into account.

To the best of our knowledge, not many researches on the WDS tank design incorporating wind power into the power system have been reported. Although larger tank size allows more water to be stored and more wind power can be accommodated, it may have adverse effect on water quality. Residual chlorine concentration is one of the most important indicators to evaluate the water quality. It should be maintained above the minimum threshold at each water node. If not, harmful microorganisms will breed and the water will not be safe anymore. Considering the interaction between WDS and power system, the optimal tank design scheme can be determined, which maximizes the deliverability of WDS energy flexibility. The upgraded WDS is a promising solution to the power system in deficiency of flexibility.

This paper proposes an optimization model of water tank size design for power system flexibility and water quality. The design model is a two-step process. The first step generates the flexibility requirements considering wind power and electrical demand uncertainties. Then the optimal tank size and the operation of WDS are determined according to the flexibility requirements and water quality criterion.

In this study, power system flexibility refers to the ability of a power system in response to the variations in electrical demand and supply due to the uncertainty of variable renewable energy. Inadequate flexibility would force the power system to

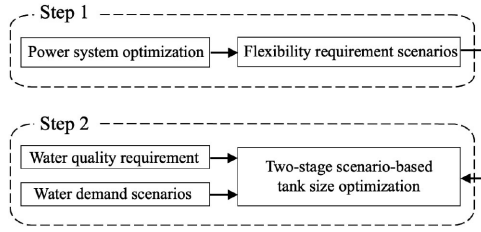


Fig. 1. Two-step optimization framework.

curtail wind power or electrical demand. If a power system lacks the upward flexibility, then the event of electrical demand curtailment will occur. Wind power curtailment will occur for a lack of downward flexibility. Flexibility requirement denotes the amount of flexibility shortage.

The water quality properties of the WDS bring nonlinear partial differential equation (NLPDE) to the model, which cannot be easily solved by the existing off-the-shelf software. To address this problem, a Benders-based heuristic algorithm is proposed to convert the NLPDE into a linear partial differential equation (LPDE). Then the WDS optimization problem is transformed into a mixed-integer programming (MIP) problem, which can be solved efficiently.

The main contributions of this paper are

- Develop an electricity-water model to leverage the energy flexibility provided by WDS and facilitate effective incorporation of wind power to the power system;
- Propose a method for optimal water tank design considering power system flexibility and water quality. A Benders-based heuristic algorithm is developed to solve the optimization problem efficiently and to protect the data privacy of each energy system.

In the remainder of this paper, the proposed optimization model is described in Section II. Section III presents the solution method using the Benders-based algorithm. Case studies are given in Section IV to verify the advantages of the proposed model. The paper is concluded in Section V.

II. THE PROPOSED OPTIMIZATION MODEL

This section presents the optimization model of water tank size design in the WDS for power system flexibility and water quality. The ultimate goal is to minimize the annual investment and operating costs. Prior to the detailed description of the proposed optimization model, the structure of the optimization model is outlined.

A. Optimization Structure

1) *Two-Step Model for Tank Size Optimization*: The WDS optimization is a two-step model for water tank size optimization. The process is depicted in Fig. 1. In the first step, flexibility requirement scenarios are generated, which reflect the upward and downward flexibility requirements of power system.

Step 2 determines the optimal tank size of WDS under the flexibility requirement scenarios, water demand scenarios, and the water quality requirement. The operation of pumps and tanks are coordinated in the WDS to adjust the electricity consumption so that the flexibility requirements of power system can be met.

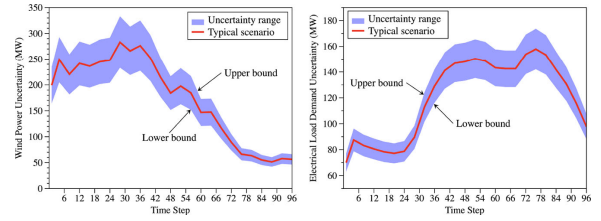


Fig. 2. Typical daily value and uncertainty range.

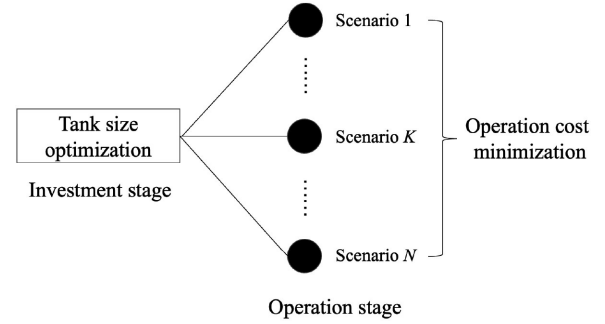


Fig. 3. Two-stage scenario-based optimization framework.

The tank size should not be too large to guarantee good water quality.

2) *Generation of Scenarios*: The seasonal impact is considered by dividing the historical data of wind power, electrical demand and water demand into four seasons with a 24-hour profile. Seasonal scenarios are then generated by using the GAMS SCENRED tool based on Backward/Forward algorithm [25]. The corresponding wind power, electrical demand and water demand can be found in each seasonal scenario.

The variations of wind power and electrical demand in real-time operation are limited by the uncertainty range, as shown in Fig. 2. The study resolution is 15 minutes in this paper, and there are 96 time steps in each scenario.

The volatility of water load is much lower than that of wind power and electrical load. The variation of water demand in real-time operation is neglected in many water tank planning researches [18]–[24]. To simplify calculations, this paper does not consider it either. In the future, research will be conducted to model the water demand variations.

A two-stage robust optimization (TSRO) method is used to find the scenarios where power system is in urgent need of flexibility. The worst-case wind power output and electrical demand in the real-time operation that jeopardize the operation cost of power system are found by considering the typical seasonal value and uncertainty range as shown in Fig. 2. In the worst-case scenario, the flexibility requirements during each time period can be determined by calculating the amount of wind power or electrical demand curtailments. Mathematical formulations of the TSRO method and detailed process of generating the flexibility requirement scenarios are given in Section II-B below.

3) *Two-Stage Scenario-Based Tank Size Optimization*: The optimal tank size is determined through a two-stage scenario-based optimization method, as seen in Fig. 3. The investment

(planning) stage optimizes the tank size considering the construction cost and engineering limitations. Once the tank size is determined, the operation stage simulates the system's daily (24-hour) operation to optimize the operation cost. In this way, planning and operation are integrated in a holistic model by taking detailed operation constraints into account in the planning stage. The two-stage scenario-based optimization ensures the investment decision to be effective and economical during the operation periods. This approach has been widely used in the field of power system planning [25]–[29].

B. Power System Optimization

The flexibility requirement is generated by feeding the typical seasonal scenarios into a power system optimization model. The objective function of the optimization model can be expressed as

$$C_E = \min_{\mathbb{X}} \left\{ C_S(\mathbb{X}) + \max_{\mathbb{U}} \min_{\mathbb{Y}} C_A(\mathbb{U}, \mathbb{Y}) \right\}. \quad (1)$$

C_E is involved with two objectives: C_S and C_A . C_S denotes the sum of the commitment, startup and shutdown cost of the conventional generating unit. C_A denotes the operation cost of power system, including the electricity production cost of the conventional unit and the penalty for curtailing wind power and electrical demands. The model intends to minimize the C_S and the maximized minimum C_A .

The outer minimization problem aims at searching for the optimal status of the commitment ($I_{s,i,t}$), startup ($U_{s,i,t}$), and shutdown ($D_{s,i,t}$) for the conventional unit. The maximization operation finds the worst-case scenario that jeopardizes the operation cost of the power system. The inner minimization is to determine the amount of wind power curtailment ($P_{s,w,t}^{wc}$), electrical demand curtailment ($P_{s,n,t}^{ec}$), and the electricity output of conventional unit ($P_{s,i,t,l}^g$).

The cost C_S can be expressed as

$$C_S(\mathbb{X}) = \sum_{s=1}^{N_s} \sum_{i=1}^{N_g} \sum_{t=1}^T \omega_s * \left(c_i * I_{s,i,t} + c_i^u * U_{s,i,t} + c_i^d * D_{s,i,t} \right). \quad (2)$$

c_i, c_i^u, c_i^d denote the commitment, start-up, and shut-down cost of unit i , respectively.

The operation cost of the power system can be formulated as

$$C_A(\mathbb{U}, \mathbb{Y}) = \sum_{s=1}^{N_s} \omega_s * \left(\sum_{w=1}^{N_W} \sum_{t=1}^T \lambda * P_{s,w,t}^{wc} + \sum_{n=1}^{N_E} \sum_{t=1}^T \mu * P_{s,n,t}^{ec} + \sum_{i=1}^{N_g} \sum_{t=1}^T \sum_{l=1}^L K_{i,l} * P_{s,i,t,l}^g \right). \quad (3)$$

The production cost of the conventional unit is expressed as a linear function $\sum_{i=1}^{N_g} \sum_{t=1}^T \sum_{l=1}^L K_{i,l} * P_{s,i,t,l}^g$ using the piecewise linearization approximation [30].

For any scenario s at time instant t , constraints of the power system are listed below.

$$I_{s,i,t} - I_{s,i,t-1} = U_{s,i,t} - D_{s,i,t} \quad \forall i \in \Omega^T \quad (4)$$

$$U_{s,i,t} + D_{s,i,t} \leq 1 \quad \forall i \in \Omega^T \quad (5)$$

$$U_{s,i,t} + \sum_{l=1}^{TU_i-1} D_{s,i,t+l} \leq 1 \quad \forall i \in \Omega^T \quad (6)$$

$$D_{s,i,t} + \sum_{l=1}^{TD_i-1} U_{s,i,t+l} \leq 1 \quad \forall i \in \Omega^T \quad (7)$$

$$\left| \sum_{l=1}^L P_{s,i,t,l}^g - \sum_{l=1}^L P_{s,i,t-1,l}^g \right| \leq Ramp_i \quad \forall i \in \Omega^T \quad (8)$$

$$P_i^{gmin} * I_{s,i,t} \leq \sum_{l=1}^L P_{s,i,t,l}^g \leq P_i^{gmax} * I_{s,i,t} \quad \forall i \in \Omega^T \quad (9)$$

$$P_{s,i,t,l}^g \geq 0 \quad \forall i \in \Omega^T, l \in \Omega^L \quad (10)$$

$$\sum_{i \in \Omega_n^T} \sum_{l=1}^L P_{s,i,t,l}^g + \sum_{w \in \Omega_n^W} (P_{s,w,t}^{pre} * R_{s,w,t}^{WN} - P_{s,w,t}^{wc}) = d_{s,n,t} * R_{s,n,t}^{EL} - P_{s,n,t}^{ec} + \sum_{h \in \Omega_n^{WDS}} P_{s,h,t}^{WDSO} + \sum_{m \in \Lambda_n^E} B_{n,m} (\theta_{s,n,t} - \theta_{s,m,t}) \quad \forall n \in \Omega^E \quad (11)$$

$$P_{s,w,t}^{pre} * R_{s,w,t}^{WN} - P_{s,w,t}^{wc} \geq 0 \quad \forall w \in \Omega^W \quad (12)$$

$$P_{s,w,t}^{wc} \geq 0 \quad \forall w \in \Omega^W \quad (13)$$

$$d_{s,n,t} * R_{s,n,t}^{EL} - P_{s,n,t}^{ec} \geq 0 \quad \forall n \in \Omega^E \quad (14)$$

$$P_{s,n,t}^{ec} \geq 0 \quad \forall n \in \Omega^E \quad (15)$$

$$R_{s,w,t}^{WN,down} \leq R_{s,w,t}^{WN} \leq R_{s,w,t}^{WN,up} \quad \forall w \in \Omega^W \quad (16)$$

$$R_{s,n,t}^{EL,down} \leq R_{s,n,t}^{EL} \leq R_{s,n,t}^{EL,up} \quad \forall n \in \Omega^E \quad (17)$$

$$-PF_{n,m} \leq B_{n,m} (\theta_{s,n,t} - \theta_{s,m,t}) \leq PF_{n,m} \quad \forall (n, m) \in \Psi^E \quad (18)$$

$$\theta_{s,ref,t} = 0. \quad (19)$$

Constraints (4) and (5) restrict the status of the generating unit. Operation limitations of the unit are described in constraints from (6) to (10). Constraint (11) is the nodal flow equation. Constraints (12), (13), (14) and (15) give the value ranges of the wind power and load curtailment. (16) describes the fluctuation range of the actual wind power, and (17) that of the actual electrical demand. The size of the uncertainty enlarges when $R_{s,w,t}^{WN,down} / R_{s,n,t}^{EL,down}$ falls and $R_{s,w,t}^{WN,up} / R_{s,n,t}^{EL,up}$ rises, indicating more conservative results. The direct current (DC) power flow is used. Constraint (18) shows the power flow limit of the line, and constraint (19) enforces the phase angle of the reference bus to zero.

The worst-case scenario occurs where the power system is most likely to face a shortage of flexible resources. The worst-case profile most likely appears after solving the three-level min-max-min optimization problem using the column-and-constraint generation (C&CG) algorithm [31]. The wind power output and electrical demand in the worst-case scenario can be calculated as $P_{s,w,t}^{pre} * \hat{R}_{s,w,t}^{WN}$ and $d_{s,n,t} * \hat{R}_{s,n,t}^{EL}$, respectively, where $\hat{R}_{s,w,t}^{WN}$, $\hat{R}_{s,n,t}^{EL}$, $\hat{P}_{s,w,t}^{wc}$, $\hat{P}_{s,n,t}^{ec}$ denote the values of variable $R_{s,w,t}^{WN}$,

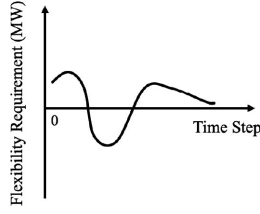


Fig. 4. Example of the flexibility requirement scenario.

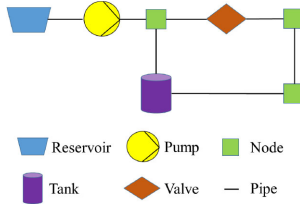


Fig. 5. Example of a WDS.

$R_{s,n,t}^{EL}$, $P_{s,w,t}^{wc}$ and $P_{s,n,t}^{ec}$, respectively. The upward flexibility requirement can be calculated by $\sum_{n=1}^{N_{EB}} \hat{P}_{s,n,t}^{ec}$ and the downward flexibility requirement by $\sum_{w=1}^{N_W} \hat{P}_{s,w,t}^{wc}$. Then the flexibility requirement scenario can be generated as shown in Fig. 4. The positive MW and the negative MW denote the upward and downward flexibility requirements, respectively. The solution procedure of C&CG will be described in Section III.

The flexibility requirement scenario is obtained based on the robust optimization method, which finds the situation where power system is most likely to face a shortage of flexible resources. Traditional robust optimization method may result in the over-conservative solutions, whereas the TSRO method used in this paper can compromise between the robustness and the optimality [31].

The unit status is optimized in the outer minimization problem of (1) (the first stage) and is considered fixed for the inner maximized minimal problem (the second stage). When the uncertainties are realized in the second stage, operation strategies of power system can be adjusted accordingly to minimize the system operation cost. Through the interaction between two stages, the TSRO method not only overcomes the overly conservative characteristic of the traditional robust optimization but also ensures the constraints to be always fulfilled. Results of the TSRO tend to be less conservative when $R_{s,w,t}^{WN,down}/R_{s,n,t}^{EL,down}$ rises and $R_{s,w,t}^{WN,up}/R_{s,n,t}^{EL,up}$ falls. TSRO degrades to a stochastic optimization problem when $R_{s,w,t}^{WN,down}$, $R_{s,n,t}^{EL,down}$, $R_{s,w,t}^{WN,up}$, and $R_{s,n,t}^{EL,up}$ equal 1.

C. WDS Optimization

1) *Overview of WDS*: The structure of a WDS is shown in Fig. 5. The reservoir serves as the source of water for WDS. The water flow is transported from one node to another in the pipe due to the head pressure difference between two nodes. The pump is used when the pressure difference is not enough. Valves control the pressure in a pipe or the amount of water flowing through

a pipe. The water tank can store water or supply water into the system.

The water pump in a WDS consumes electricity to increase the pressure difference and transport the water flow. More water can be conveyed when the electricity consumption is increased. The water can be stored in the tank to ensure the balance between supply and demand. Thus, WDS can be regarded as an adjustable electrical load from the perspective of power system. Optimal control of the water pump and tank enables WDS to deal with the flexibility requirements of power system and to incorporate more wind power.

2) *Tank Size Optimization*: Now we describe the two-stage scenario-based tank size optimization problem involved with WDS. The objective function can be formulated as (20) shown at bottom of the next page, where the first stage of the objective function includes the tank construction cost. The second stage includes the electricity bill and the penalty for not satisfying the flexibility requirements. The value of $P_{s,t}^{Flex}$ depends on the flexibility requirement scenario generated from power system optimization problem. The gap between electricity consumption adjustment of WDS and power system flexibility requirement can be calculated by

$$\sum_{t=1}^T \left(\left(\sum_{h=1}^{N_{WDS}} P_{s,h,t}^{WDS} - P_{s,h,t}^{WDSO} \right) + P_{s,t}^{Flex} \right)^2.$$

The penalty factor ξ ensures that the power system flexibility requirement be satisfied as much as possible.

For any scenario s and time t , the hydraulic constraints of WDS are given from (21) to (36).

$$h_{s,J_m}^t - h_{s,J_n}^t = c_{mn} * \text{sign}(q_{s,mn}^t) * (q_{s,mn}^t)^2 \quad \forall (J_m, J_n) \in \Delta \quad (21)$$

$$q_{mn}^{min} \leq q_{s,mn}^t \leq q_{mn}^{max} \quad \forall (J_m, J_n) \in \Delta \quad (22)$$

$$\text{Pump on} : \hat{h}_{s,P_m}^t = a_{P_m} (q_{s,P_m}^t)^2 + b_{P_m} q_{s,P_m}^t + c_{P_m} \quad q_{P_m}^{min} \leq q_{s,P_m}^t \leq q_{P_m}^{max} \quad \forall P_m \in \Delta_P \quad (23)$$

$$\text{Pump off} : q_{s,P_m}^t = 0 \quad \forall P_m \in \Delta_P \quad (24)$$

$$-M(1-z_{s,mn}^t) \leq h_{s,J_n}^t - h_{s,J_m}^t + R_{s,mn}^t \leq M(1-z_{s,mn}^t) \quad \forall (J_m, J_n) \in \Delta_V \quad (25)$$

$$R_{s,mn}^t \geq 0 \quad \forall (J_m, J_n) \in \Delta_V \quad (26)$$

$$q_{mn}^{min} * z_{s,mn}^t \leq q_{s,mn}^t \leq q_{mn}^{max} * z_{s,mn}^t \quad \forall (J_m, J_n) \in \Delta_V \quad (27)$$

$$dw_{s,J_k}^t = \sum_{J_n:(J_n,J_k) \in \Delta} q_{s,nk}^t - \sum_{J_m:(J_k,J_m) \in \Delta} q_{s,km}^t \quad \forall J_k \in \varpi \quad (28)$$

$$h_{s,J_m}^t \geq h_{J_m}^{min} \quad \forall J_m \in \varpi \quad (29)$$

$$q_{s,J_m,t}^{in} + q_{s,J_m,t}^{out} = (h_{s,J_m}^{t+1} - h_{s,J_m}^t) * A_{J_m}^{tank} \quad \forall J_m \in \varpi_T \quad (30)$$

$$W_{s,J_m,t} = (1 - \mu_W) W_{s,J_m,t-1} + q_{s,J_m,t-1}^{in} + q_{s,J_m,t-1}^{out} \quad \forall J_m \in \varpi_T \quad (31)$$

$$W_{J_m}^{min} \leq W_{s,J_m,t} \leq W V_h^{J_m} \quad \forall J_m \in \varpi_T \quad (32)$$

$$W V_h^{J_m} = W_{J_m}^{guide} + W_{J_m}^{wind} \quad \forall J_m \in \varpi_T \quad (33)$$

$$W V_h^{J_m} \leq W V_h^{max} \quad \forall J_m \in \varpi_T \quad (34)$$

$$P_{s,i,t}^{WP} = \kappa_1 (q_{s,P_i}^t)^3 + \kappa_2 (q_{s,P_i}^t)^2 + \kappa_3 q_{s,P_i}^t \quad \forall P_i \in \Delta_P \quad (35)$$

$$P_{s,h,t}^{WDS} = \sum_{i=1}^{N_h^{WP}} P_{s,i,t}^{WP} \quad \forall h \in \Omega^{WDS} \quad (36)$$

Constraint (21) describes the relationship between the head loss and the flow in a pipe using the Darcy-Weisbach equation. Sign function $sign(q_{s,mn}^t)$ ensures that the pressure is lowered in the flow direction. Constraint (22) limits the water volume in the pipe. The fixed-speed pump is used and modeled by (23) and (24). The pressure reducing valve is adopted, as formulated by (25), (26) and (27). Equation (28) describes the mass balance, and inequality (29) restricts the pressure head of each node. The water tank is modeled by constraints from (30) to (34). According to governmental guidelines, the tank size should satisfy several usage requirements such as the balancing storage and the reserve [17]. $W_{J_m}^{guide}$ is the required constructed size to meet guidelines. The electricity consumption of the pump can be obtained by (35). The total electricity consumption of WDS is the sum of all the pumps, as seen in (36).

Water quality can be described by the residual chlorine concentration C_{mn} , which can be obtained by the water quality equations from (37) to (42). $C_{mn}(s, x, t)$ is a function of the scenario s , segment x , and time t .

$$\frac{\partial C_{mn}(s, x, t)}{\partial t} + v_{mn}(s, t) \frac{\partial C_{mn}(s, x, t)}{\partial x} + r [C_{mn}(s, x, t)] = 0 \quad \forall (J_m, J_n) \in \Delta \quad (37)$$

$$v_{mn}(s, t) = q_{s,mn}^t / S_{mn} \quad (38)$$

$$r [C_{mn}(s, x, t)] = k * C_{mn}(s, x, t) \quad (39)$$

$$C_{J_m}(s, t) = \frac{\left(\sum_{(J_n, J_m) \in \Delta^+} q_{s,nm}^t * C_{nm}(s, L_{nm}, t) + Q_{s,J_m,ex}^t * C_{J_m,ex}(s, t) \right)}{\sum_{(J_n, J_m) \in \Delta^+} q_{s,nm}^t + Q_{s,J_m,ex}^t} \quad \forall J_m \in \varpi \quad (40)$$

$$W_{s,J_m,t} * C_{J_m}(s, t) - W_{s,J_m,t-\Delta t} * C_{J_m}(s, t - \Delta t) = \Delta t * \left(\sum_{(J_n, J_m) \in \Delta^+} q_{s,nm}^{t-\Delta t} * C_{nm}(s, L_{nm}, t - \Delta t) - \sum_{(J_m, J_k) \in \Delta^+} q_{s,mk}^{t-\Delta t} * C_{J_m}(s, t - \Delta t) - r [C_{J_m}(s, t - \Delta t)] \right) \quad \forall J_m \in \varpi_T \quad (41)$$

$$r [C_{J_m}(s, t)] = k_b * C_{J_m}(s, t). \quad (42)$$

The conservation of mass during transport along the pipe is described by the one-dimensional advection equation of (37). The flow velocity is calculated by (38), and the reaction rate of pipe is calculated by (39). The balancing constraint for water nodes is described by (40). The tank is assumed to be mixed and (41) describes the balancing constraint. The tank reaction rate is calculated by (42). For pumps and valves in the WDS, the substance passes through instantaneously and the concentration remains unchanged.

The electricity consumption scheme of the WDS considering tank size optimization should ensure that the power flow limit of the transmission line in the power system is not violated. Thus the following constraints should be obeyed:

$$\sum_{i \in \Omega_n^T} \left(\sum_{l=1}^L \hat{P}_{s,i,t,l}^g \right) + \sum_{w \in \Omega_n^W} \left(P_{s,w,t}^{pre} * \hat{R}_{s,w,t}^{WN} - P_{s,w,t}^{wca} \right) = d_{s,n,t} * \hat{R}_{s,n,t}^{EL} - P_{s,n,t}^{eca} + \sum_{h \in \Omega_n^{WDS}} P_{s,h,t}^{WDS} + \sum_{m \in \Lambda_n^E} B_{n,m} (\theta_{s,n,t}^{AJ} - \theta_{s,m,t}^{AJ}) \quad \forall n \in \Lambda^E - P_{F,n,m} \leq B_{n,m} (\theta_{s,n,t}^{AJ} - \theta_{s,m,t}^{AJ}) \leq P_{F,n,m} \quad \forall (n, m) \in \Psi^E \quad (43)$$

$$\theta_{s,ref,t}^{AJ} = 0 \quad (45)$$

All the symbols in the form of $\widehat{(\cdot)}$ denote the values of the corresponding variables $(\cdot)$ in Step 1 of the power system optimization problem. The optimal tank size can be determined by solving the optimization problem with the objective function (20) and constraints from (21) to (45).

The flexibility requirement after the tank size optimization can be calculated by

$$P_{s,t}^{Flexa} = \sum_{n=1}^{N_{EB}} \hat{P}_{s,n,t}^{eca} - \sum_{w=1}^{N_W} \hat{P}_{s,w,t}^{wca} \quad (46)$$

where $\hat{P}_{s,n,t}^{eca}$ and $\hat{P}_{s,w,t}^{wca}$ denote the values of $P_{s,n,t}^{eca}$ and $P_{s,w,t}^{wca}$, respectively.

$$C_{WDS}(\mathbb{Z}) = \min \left(\sum_{h=1}^{N_{WDS}} \sum_{n=1}^{N_h^{WT}} W V_h^n * \varpi_h^n * \frac{r(1+r)^{IY}}{(1+r)^{IY}-1} + \sum_{s=1}^{N_{sc}} \omega_s * \left(\sum_{t=1}^T \left(\sum_{h=1}^{N_{WDS}} P_{s,h,t}^{WDS} * \lambda price_t + \xi * \sum_{t=1}^T \left(\sum_{h=1}^{N_{WDS}} P_{s,h,t}^{WDS} - P_{s,h,t}^{WDSO} \right) \right)^2 \right) \right) \quad (20)$$

D. Supplementary Remarks

$P_{s,h,t}^{WDSO}$ denotes the electrical demand used by WDS without optimizing the tank size. It serves as the input parameter in (11), (20). It can be determined as the value of variable $P_{s,h,t}^{WDS}$ after solving the optimization problem

$$\begin{cases} \min \left(\sum_{s=1}^{N_{sc}} \omega_s^* \sum_{h=1}^{N_{WDS}} \sum_{t=1}^T P_{s,h,t}^{WDS} * \lambda_{price_t} \right) \\ \text{s.t. (21) - (31), (35) - (36)} \\ W_{J_m}^{min} \leq W_{s,J_m,t} \leq W_{J_m}^{guide} \end{cases} \quad (47)$$

Note that the optimization aims at minimizing the electricity bill of WDS without providing flexibility for power system.

III. SOLUTION METHODOLOGY

Many commercial solvers such as Gurobi are available for power system solutions. But they are not applicable to the min-max-min optimization problem of (1) and the nonlinear partial differential equation (NLPDE) in the water distribution system (WDS) model [32]. Thus, we apply the C&CG algorithm and propose a Benders-based heuristic algorithm to transform the min-max-min optimization problem of (1) and NLPDE into the MIP problems so that the commercially available solvers can be applied. Details are given below.

A. Solution Procedure of C&CG Algorithm

The min-max-min optimization problem of the power system can be expressed in the following form

$$\begin{cases} C_E = \min_{\mathbb{X}} \left\{ C_S(\mathbb{X}) + \max_{\mathbb{U}} \min_{\mathbb{Y}} C_A(\mathbb{U}, \mathbb{Y}) \right\} \\ \text{s.t. } F(\mathbb{X}) \leq D \\ G(\mathbb{X}) + H(\mathbb{Y}) + J(\mathbb{U}) \leq R. \end{cases} \quad (48)$$

The C&CG exploits a master-slave framework formulated as

$$\begin{cases} \min_{\mathbb{X}, \lambda} C_S(\mathbb{X}) + \lambda \\ \text{s.t. } F(\mathbb{X}) \leq D \\ \lambda \geq C_A(\mathbb{U}^*, \mathbb{Y}) \\ G(\mathbb{X}) + H(\mathbb{Y}) + J(\mathbb{U}^*) \leq R \quad \forall \mathbb{U}^* \in Sl \end{cases} \quad (49)$$

and

$$\begin{cases} \max_{\mathbb{U}} \min_{\mathbb{Y}} C_A(\mathbb{U}, \mathbb{Y}) \\ \text{s.t. } G(\mathbb{X}^*) + H(\mathbb{Y}) + J(\mathbb{U}) \leq R. \end{cases} \quad (50)$$

The master-slave problem framework of C&CG enables system operators to find the amount of wind power curtailment (i.e., values of $P_{s,w,t}^{wc}$) and the amount of electrical demand curtailment (i.e., values of $P_{s,n,t}^{ec}$) under the worst-case wind power output and electrical demand.

In the master problem (49), an optimal solution $(\mathbb{X}^*, \lambda^*)$ can be obtained with a set of the fixed variables in $\mathbb{U}^*$ generated from the slave problem. The slave problem then solves (50) based on the obtained $\mathbb{X}^*$. (50) can be considered as a bi-level problem, which is solved using the KKT conditions and Big-M approach [31]. Solution details of (50) are given in the Appendix A. The solution procedure of C&CG is summarized in Algorithm 1. The obtained optimal solution $\mathbb{U}^*$ in the last iteration collects the

Algorithm 1: C&CG.

-
- Step 1 Initialize the lower bound LB as $-\infty$ and the upper bound UB as $+\infty$, the slave solution set Sl as $\emptyset$ and the iteration number i as 0.
- Step 2 Solve the master problem (49) and update $\mathbb{X}_{i+1}^*$, λ_{i+1}^* , and lower bound $LB = \max\{LB, C_S(\mathbb{X}_{i+1}^*) + \lambda_{i+1}^*\}$.
- Step 3 Solve the slave problem (50) to obtain the optimal solution $\mathbb{U}^*$, add to the set Sl . Update the upper bound $UB = \min\{UB, C_S(\mathbb{X}_{i+1}^*) + \lambda_{i+1}^* + C_A(\mathbb{U}_{i+1}^*, \mathbb{Y}_{i+1}^*)\}$.
- Step 4 If the convergence criterion $(UB - LB) \leq \epsilon_1$ is met, then Stop; otherwise, $i = i + 1$ and go to Step 2.
-

worst-case wind power and electrical demand curve of each daily scenario, which is used to generate the flexibility requirement scenario in Section II.

B. Benders Decomposition

The WDS optimization problem is formulated as

$$\begin{cases} \min C_{WDS}(\mathbb{Z}) \\ \text{s.t. (21) - (45)}. \end{cases} \quad (51)$$

$C_{WDS}(\mathbb{Z})$ denotes the total cost of WDS with the optimized tank size, as described in (20). It can be decomposed into a master problem and a slave problem by the Benders decomposition.

1) Master problem:

$$\begin{cases} \min C_{WDS}(\mathbb{Z}) \\ \text{s.t. (21) - (42)}. \end{cases} \quad (52)$$

The master problem is solved with the data from WDS. The obtained values of $P_{s,h,t}^{WDS}$ are then passed to the slave problem.

1) Slave problem:

The slave problem can be formulated as

$$\begin{aligned} & \min \left(\sum_{s=1}^{N_{sc}} \sum_{n=1}^{N_{EB}} \sum_{t=1}^T (\xi_{s,n,t}^{P1} + \xi_{s,n,t}^{P2}) \right) \\ & \text{s.t. (44) - (45);} \\ & P_{s,h,t}^{WDS} = \hat{P}_{s,h,t}^{WDS}(\gamma_{s,h,t}); \end{aligned} \quad (53)$$

$$\sum_{i \in \Omega_n^T} \left(\sum_{l=1}^L \hat{P}_{s,i,t,l}^g \right) + \sum_{w \in \Omega_n^W} (P_{s,w,t}^{pre} * \hat{R}_{s,w,t}^{WN} - P_{s,w,t}^{wca}) + \xi_{s,n,t}^{P1} \quad (54)$$

$$\begin{aligned} & = \xi_{s,n,t}^{P2} + d_{s,n,t} * \hat{R}_{s,n,t}^{EL} - P_{s,n,t}^{eca} + \sum_{h \in \Omega_n^{WDS}} P_{s,h,t}^{WDS} \\ & + \sum_{m \in \Lambda_n^E} B_{n,m} (\theta_{s,n,t}^{AJ} - \theta_{s,m,t}^{AJ}) \quad \forall n \in \Lambda^E. \end{aligned}$$

The nodal power balancing constraint (43) is converted into equations (53) and (54), because the obtained values of $P_{s,h,t}^{WDS}$ from the master problem may make it obsolete. Equation (53)

receives the values of $P_{s,h,t}^{WDS}$ (i.e., $\hat{P}_{s,h,t}^{WDS}$) from the master problem, and equation (54) introduces non-negative slack variables $\xi_{s,n,t}^{P1}$ and $\xi_{s,n,t}^{P2}$ to ensure the balance between supply and demand. $\gamma_{s,h,t}$ is the dual variable for the constraint (53).

If the objective function of the slave problem is equal to zero, the iterative process ends. Otherwise, an optimality cutting plane (55) is generated as a new constraint to the master problem. This constraint can be expressed as

$$\begin{aligned} & \sum_{s=1}^{N_{sc}} \sum_{n=1}^{N_{EB}} \sum_{t=1}^T \left(\hat{\xi}_{s,n,t}^{P1} + \hat{\xi}_{s,n,t}^{P2} \right) \\ & + \sum_{s=1}^{N_{sc}} \sum_{h=1}^{N_{WDS}} \sum_{t=1}^T \gamma_{s,h,t}^k \left(P_{s,h,t}^{WDS} - \hat{P}_{s,h,t}^{WDS} \right) \leq 0 \end{aligned} \quad (55)$$

where $\hat{\xi}_{s,n,t}^{P1}$ denotes the value of $\xi_{s,n,t}^{P1}$, $\hat{\xi}_{s,n,t}^{P2}$ the value of $\xi_{s,n,t}^{P2}$, and $\gamma_{s,h,t}^k$ the value of $\gamma_{s,h,t}$ at the k^{th} iteration.

Note that the feasibility cutting plane is not required in the master problem because the slave problem is always feasible with the introduction of slack variables. This “trick” has proven to be computationally efficient [33].

Thus, the WDS optimization can be solved iteratively between the master problem and the slave problem till the termination criterion is met.

C. Proposed Benders-Based Heuristic Algorithm

The sign function in (21) and the “on/off” statement in (23) and (24) need to be further transformed, as shown in the Appendix B. Then the master problem (52) is turned into:

$$\begin{cases} \min C_{WDS}(\mathbb{Z}) \\ \text{s.t. (22), (25) - (42), (B1) - (B8)} \end{cases} \quad (56)$$

A Benders-based heuristic algorithm (Algorithm 2) is proposed to make the water quality equations (37)-(42) more tractable. The master problem (56) can be solved after several iterations between Step 2 and Step 3. Small φ affects the optimality of the problem, whereas large φ may increase the number of iterations. The slave problem is a linear programming (LP) model, which ensures the convergence between master and slave problems in Benders decomposition [34]–[35]. Thus, convergence of the proposed Benders-based heuristic algorithm is guaranteed.

A constraint on the tank size (57) is added to make solutions of the optimization problem (56) more feasible to the real world applications. The lower bound of the tank size is limited by the government guidelines and the upper bound is governed by (57)

$$W_{J_n}^{guide} \leq W V_h^n \leq \varphi \widehat{W V}_h^n.$$

The purpose of the upper bound is to avoid the situation that the oversized tank may jeopardize the water quality.

The NLPDE of (37) becomes a linear equation with the following process. Divide the pipe into multiple segments uniformly and apply the backward difference of (58) and (59) with the values of $q_{s,mn}^t$ and $W_{s,J_m,t}$ obtained from (56). Equations (40) and (41) also become linear. The variable x in (58) and (59)

Algorithm 2: Benders-Based Heuristic Algorithm.

Step 1 Initialize the number of iteration $k = 0$.

Step 2 Solve the master problem (56).

Step 3 Apply the values of $q_{s,mn}^t$ and $W_{s,J_m,t}$ obtained from (56) to the water quality equations (37)-(42). If the water quality is acceptable, pass the values of $P_{s,h,t}^{WDS}$ to the slave problem and go to Step 4; otherwise, add the constraint (57) to (56) and go to Step 2.

$$W V_h^n \leq \varphi \widehat{W V}_h^n. \quad (57)$$

$\widehat{W V}_h^n$ is the value of $W V_h^n$ obtained from Step 2, and φ is a parameter with value in the range of [01].

Step 4 Solve the slave problem. If the value of the objective function in the slave problem is equal to zero, the iterative process is terminated; otherwise, a cutting plane constraint (55) should be included in the master problem (56). Update $k = k+1$ and return to Step 2.

refers to the x^{th} segment of the pipe.

$$\frac{\partial C_{mn}(s, x, t)}{\partial t} = \frac{C_{mn}(s, x, t) - C_{mn}(s, x, t-1)}{\Delta t} \quad (58)$$

$$\frac{\partial C_{mn}(s, x, t)}{\partial x} = \frac{C_{mn}(s, x, t) - C_{mn}(s, x-1, t)}{\Delta x}. \quad (59)$$

The flowchart of the solution process is presented in Fig. 6. The C&CG algorithm is performed by the power system operator (PSO). The flexibility requirement scenarios generated from the C&CG algorithm serve as input parameters for the Benders-based heuristic algorithm.

In the Benders-based algorithm, the master problem is solved by the water distribution system operator (WDSO) and the slave problem by PSO.

The information interchange between two systems includes: 1) WDSO passes the values of $P_{s,h,t}^{WDS}$ to PSO; 2) PSO generates the cutting plane constraint (55) and passes it to WDSO. Limited information interchange in the solution process can protect the privacy of each energy system.

IV. CASE STUDY

The proposed approach for tank size design is evaluated in this section. The tests were carried out by a PC with Intel Core i7-9700 and Gurobi 9.0.1 on Python 3.7. The time horizon of each scenario is 24 hours and the study resolution is 15 minutes. This corresponds to 96 time steps.

A. Illustrative Test System

Firstly, a relatively small test system is used to illustrate the advantages of the tank design model and the proposed algorithm

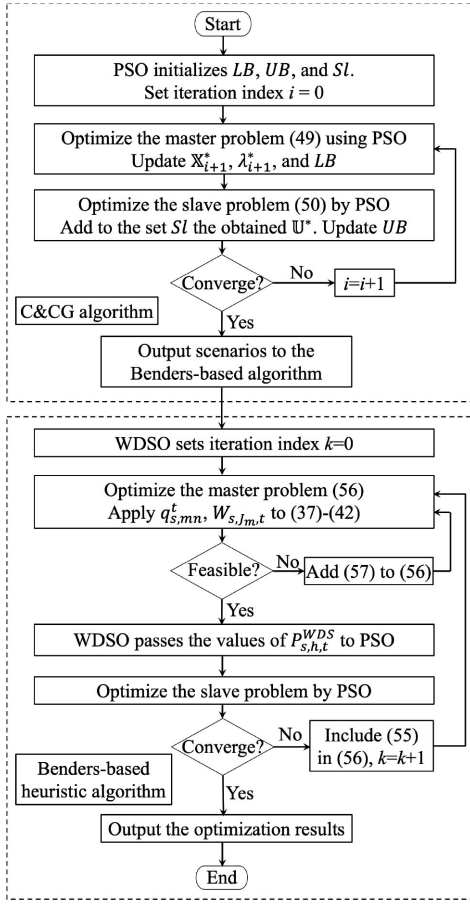


Fig. 6. Flowchart of the solution process.

TABLE I
ELECTRICITY PRICE OF WDS

	Valley	Off-peak	Peak
Time period (h)	0-8	12-17 & 21-24	8-12 & 17-21
Price (\$/MWh)	40	70	132

in detail. The test system includes the IEEE-RTS 24 power system and the WDS with 40 pipes [5].

The diagram of the IEEE-RTS 24 system is shown in Appendix C. There are 17 load buses in the IEEE-RTS 24 system and each load bus is connected with a WDS. The wind farms are located at nodes 3, 5 and 20 of the power system. The tank size should be at least 3000 m³ according to the governmental requirements. The minimum required residual chlorine at all demand nodes is set to be 0.3 mg/L. The residual chlorine concentration at the reservoir is fixed as 1.0 mg/L, and the initial concentrations at the water nodes equal 0.

The electricity price for the WDS is given in Table I. Four typical scenarios (representing the four seasons, respectively) are generated based on the scaled data of the Irish grid [36] and the WDS from Cherry Hills, Connecticut [37]. The occurrence probability of each scenario equals to 0.25. Fig. 7 illustrates the daily wind power outputs of four seasonal scenarios for three different wind farms. Fig. 8 shows the daily electrical and water load demands of four seasonal scenarios. The fluctuation range

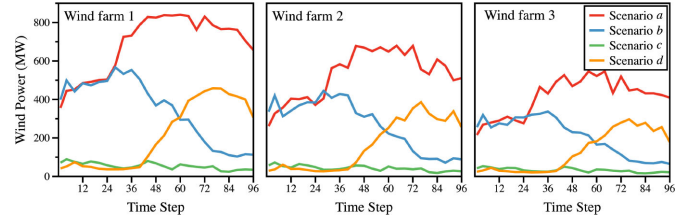


Fig. 7. Daily wind power outputs for various scenarios at three different wind farms.

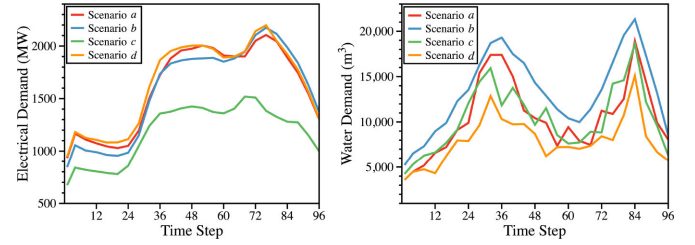


Fig. 8. Daily electrical and water load demands for various scenarios.

TABLE II
DEFINITION OF SITUATIONS AND SCENARIOS FOR EVALUATING THE PROPOSED TANK SIZE DESIGN

	Index	Definition
Situation	1	Power system
	2	Power system + WDS
	3	Power system + WDS + tank design
Scenario	a	Spring (March to May)
	b	Summer (June to August)
	c	Autumn (September to November)
	d	Winter (December to February)

of the wind power in real-time operation is set to be 75% to 125% of the values in the scenario, and that of the electrical demand is set to be 95% to 105%.

1) *Power System Flexibility Improvement*: In the case study, three situations are considered to demonstrate the benefits of the proposed tank size design scheme and to show the improvement of power system flexibility.

Situation 1: Consider the power system alone.

Situation 2: Consider power system and WDS without the proposed tank size design scheme.

Situation 3: Consider power system and WDS with the proposed tank size design scheme.

In each situation, four scenarios are considered for seasonal impact. Definitions of the situation and scenario are tabulated in Table II. Scenario *a* denotes the months from March to May, Scenario *b* from June to August, Scenario *c* from September to November, and Scenario *d* from December to February. Table III shows the upward and downward flexibility requirements for each scenario under different situations. It is observed from Table III that the largest flexibility requirement occurs at Situation 1, whereas Situation 3 requires the least flexibility. Since the wind power output generally is at a low level in the three months from September to November, there are no flexibility

TABLE III
FLEXIBILITY REQUIREMENTS IN THE ILLUSTRATIVE TEST SYSTEM

Situation	Scenario	Flexibility Requirement (MW)	
		Upward	Downward
1	<i>a</i>	1,206.38	15,603.74
	<i>b</i>	749.65	10,429.31
	<i>c</i>	118.04	503.22
	<i>d</i>	648.72	6,472.85
2	<i>a</i>	97.33	7,081.66
	<i>b</i>	32.60	1,510.23
	<i>c</i>	0	0
	<i>d</i>	0	538.42
3	<i>a</i>	0	2,562.80
	<i>b</i>	0	322.19
	<i>c</i>	0	0
	<i>d</i>	0	0

TABLE IV
COMPARISON OF THE AS COST IN THE ILLUSTRATIVE TEST SYSTEM

Situation	Scenario	AS cost (\$)
1	<i>a</i>	110,400
	<i>b</i>	65,124
	<i>c</i>	2,346
	<i>d</i>	24,869
2	<i>a</i>	34,354
	<i>b</i>	7,272
	<i>c</i>	0
	<i>d</i>	2,221
3	<i>a</i>	9,064
	<i>b</i>	1,338
	<i>c</i>	0
	<i>d</i>	0

requirements of Scenario *c* in Situations 2 and 3. Thus, variations in real-time operation can be compensated by regulating the electricity consumption of WDS. It also can be seen from Table III that, after integrating WDS into the power system, the lack of upward flexibility only occurs in Scenarios *a* and *b* of Situation 2. The event of upward flexibility shortage occurs because the electrical demand reaches its daily peak value. The sudden decrease of wind power forces the power system to curtail the demand. After applying the proposed tank size design scheme, the lack of upward flexibility can be avoided. This shows the merits of WDS and the proposed tank size design scheme in dealing with the wind power uncertainties.

The quantitatively measured economic values provided by the WDS are summarized in Table IV. These measures satisfy the flexibility requirements. The results show the auxiliary service (AS) purchased from the electricity market under various situations and scenarios. Fig. 9 displays the data used in the measurement, which are the CAISO's day-ahead forecasted frequency regulation prices for March 8, 2021 [38]. Table IV compares the cost of buying AS in the illustrative system. It is seen that the AS cost can be reduced greatly with the optimized tank size and WDS operation. This demonstrates the economic value of the proposed model.

Table V shows the tank sizes for various WDSs after WDS optimization. The tank sizes for all the WDSs equal 3000 m³ in Situations 1 and 2, which satisfy the governmental requirements. In Situation 3, the optimized tank size of each WDS

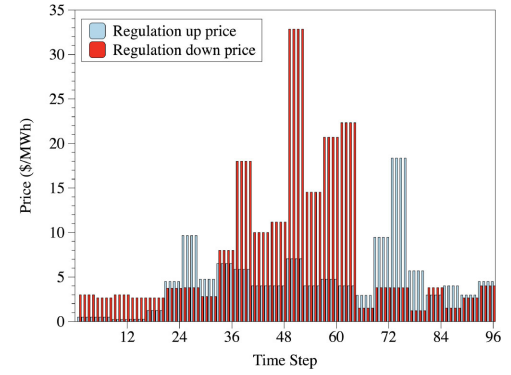


Fig. 9. Regulation prices.

TABLE V
TANK SIZE IN DIFFERENT SITUATIONS

Situation	WDS	Tank size (m ³)
1 & 2		3,000
3	1-7, 16-17	5,738 - 6,400
	8-12	4,312 - 5,062
	13-15	3,375 - 3,886

may vary because WDSs have varied connection relationships with the power system and the electrical transmission limits are considered.

In the test, 17 WDSs were divided into three categories according to the obtained optimal tank size. WDS 1-7 and 16-17 require larger tank sizes, because these WDSs are closer to the wind farms. On the other hand, since the water tank is mainly used to satisfy the water demands, not providing flexibility to the power system, additional volumes of water tank do not provide additional flexibility for WDS 13-15. These tanks are far away from the wind farms so that the variations of their electricity consumption have little effect on the wind power accommodation due to the power flow limit of the electrical transmission line.

2) *Water Quality*: The water quality also can be maintained after optimization by incorporating the water quality constraints into the WDS model. This water quality effect can be illustrated using the WDS of 40 pipes [5]. Three representative water nodes are considered: Node 2 at the beginning, Node 12 in the middle, and Node 30 at the end of the WDS. Fig. 10 shows variations of the residual chlorine concentration at these three water nodes in WDS 1 under the four scenarios. The residual chlorine concentrations at these nodes satisfy the minimum requirement of 0.3 mg/L. It is observed that Node 30 exhibits the smallest concentration, because it is located at the end of WDS, at which the water demand is low.

3) *Algorithm Performance*: The proposed Benders-based heuristic algorithm transforms the original model with NLPDE into a MIP master problem and a LP slave problem, which can be solved by PSO and WDSO, respectively. Table VI presents the performance of the convergence process of the proposed algorithm. The first column indicates the number of runs between the master problem (MP) and the slave problem (SP). The second

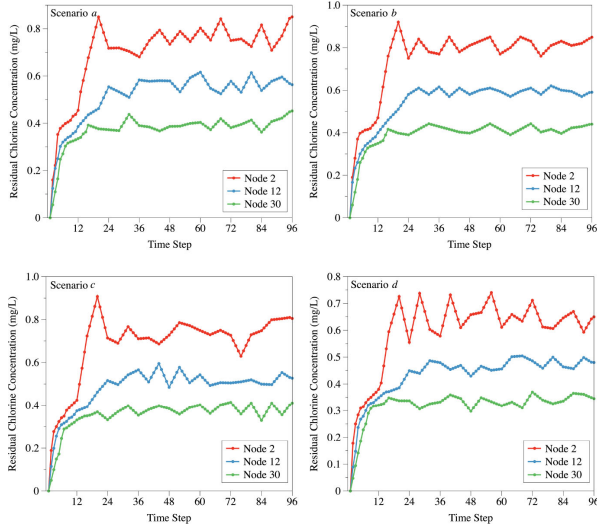


Fig. 10. Residual chlorine concentrations at three water nodes of a illustrative WDS under different scenarios. Node 30 is at the end of WDS.

TABLE VI

PERFORMANCE OF THE PROPOSED BENDERS-BASED ALGORITHM IN THE ILLUSTRATIVE TEST SYSTEM

Number of Runs	Iterations in MP	Total cost (\$)	Time (min)
1	4	568,798	10.23
2	2	568,895	6.31
3	3	569,151	7.40
4	4	569,422	9.56
5	3	569,799	7.05

TABLE VII

THE INFLUENCE OF PENALTY FACTOR ON CONVERGENCE

φ	Number of Runs	Iterations in MP	Total cost (\$)	Time (min)
0.8	6	9	582,048	21.38
0.85	7	11	577,650	25.16
0.9	5	14	571,784	34.13
0.95	4	16	569,799	40.55
0.98	5	25	569,060	67.19
0.99	7	28	568,132	85.30

column displays the total iterations required in the MP by use of the added constraint (57) on the tank size.

It is seen from Table VI that the proposed algorithm converges after 5 runs between the MP and SP. It takes no more than 4 iterations for the convergence of MP at each run. The total computing time is 40.55 minutes. Since this paper aims at the year-ahead planning, this calculation time is short enough for practical use.

According to the findings of extensive simulations, the settings of the penalty factor φ may influence the convergence efficiency of the proposed algorithm. Table VII shows the variations of the optimization results with φ .

It can be seen from Table VII that total iterations in MP and the computing time increase as φ increases. A small φ lowers the optimized tank size. Thus, the available flexibility is limited, resulting in higher cost. This study suggests that a suitable value of φ is 0.95.

TABLE VIII
FLEXIBILITY REQUIREMENTS IN THE LARGER-SCALE TEST SYSTEM

Situation	Scenario	Flexibility Requirement (MW)	
		Upward	Downward
1	a	2,743.75	34,149.83
	b	1,698.10	22,984.21
	c	1,404.41	19,830.28
	d	236.68	1,137.66
2	a	718.14	15,763.27
	b	549.08	8,104.14
	c	71.91	6,173.74
	d	0	0
3	a	206.35	5,781.24
	b	0	720.10
	c	0	367.91
	d	0	0

TABLE IX

COMPARISON OF THE AS COST IN THE LARGER-SCALE TEST SYSTEM

Situation	Scenario	AS cost (\$)
1	a	150,835
	b	117,466
	c	89,568
	d	6,145
2	a	72,082
	b	41,128
	c	32,558
	d	0
3	a	29,553
	b	3,101
	c	1,700
	d	0

B. Larger-Scale Test System

A realistic power system in a province of Northwest China is used to verify the superiority of the proposed tank size design. The power system has 93 load buses and 179 transmission lines. The installed wind power capacity is 3667.2 MW, accounting for 12% of the total generating capacity. The “D-Town” WDS with 443 pipes is used [39], which often serves as benchmarks for water network design methods. Each load bus in the power system is connected with a WDS. Four typical seasonal scenarios are generated using the GAMS SCENRED tool, and the occurrence probability of each scenario equals to 0.25. The tank size should be at least 3500 m³ according to the governmental requirements. The water quality requirements, initial conditions of residual chlorine, and the fluctuation ranges of wind power and electrical demand are the same as the test system in Section A.

1) *Power System Flexibility Improvement*: The three situations referred in Section A are considered again. Table VIII shows the flexibility requirements in these three situations. Comparison of the AS cost is displayed in Table IX. It can be seen from Table VIII that both the upward and downward flexibility requirements can be reduced greatly with the proposed tank size design scheme. The AS cost in each scenario is reduced by about 80% to 100% after the flexible operation of WDS with the optimized tank size. This validates the economic merit of the proposed model.

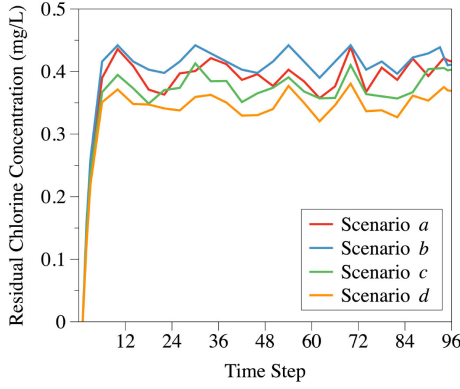


Fig. 11. Residual chlorine concentration at Node 399, the end of a larger-scale WDS.

TABLE X
PERFORMANCE OF THE PROPOSED BENDERS-BASED ALGORITHM IN THE LARGER-SCALE TEST SYSTEM

Number of Runs	Iterations in MP	Total cost (\$)	Time (min)
1	6	1,121,837	34.70
2	5	1,259,809	29.17
3	3	1,277,226	19.01
4	6	1,303,680	37.90
5	6	1,347,491	36.10
6	4	1,353,683	25.31
7	2	1,370,482	13.04
8	3	1,374,514	20.54
9	5	1,376,540	27.74

2) *Water Quality*: The water quality of Node 399, which is at the end of the WDS, is the worst. Fig. 11 shows variations of the residual chlorine concentration at Node 399 under the four scenarios. The findings demonstrate that the proposed WDS model can guarantee the required water quality.

3) *Algorithm Performance*: The penalty factor φ is set as 0.95. Table X shows the convergence process of the proposed algorithm. It can be seen from Table X that it only takes 243.51 minutes for convergence. This length of time is acceptable for practical use.

V. CONCLUSION

This paper has presented an optimization model of WDS to enhance the power system flexibility and incorporate more wind power. The proposed model generates the flexibility requirements considering wind power and electrical demand uncertainties, then finds the optimal size of water tank based on the flexibility and water quality requirements. A Benders-based heuristic algorithm is developed to facilitate the solutions with a master-slave framework. The master problem is solved in the WDS setting and the slave problem via the power system. Since only limited information needs to be shared, data privacy can be protected.

Results from case studies have verified that the optimized tank size can enhance the flexibility of the power system. Both the amount of wind power curtailment and that of the electrical demand curtailment are reduced significantly. Moreover, the

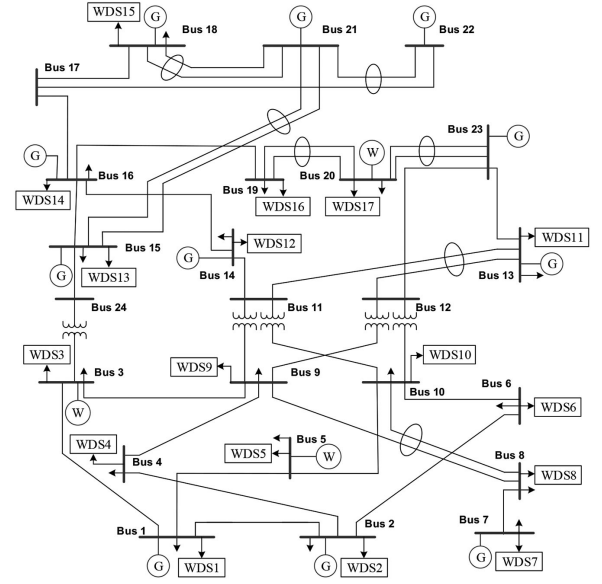


Fig. 12. Diagram of the IEEE-RTS 24 power system.

proposed model maintains the required water quality at all water nodes of the WDS.

APPENDIX

A. Solution Details of (50)

By applying KKT conditions, the max-min optimization problem (50) can be transformed into

$$\begin{aligned}
 & \max_{\mathbf{U}} C_A(\mathbf{U}, \mathbf{Y}) \\
 & \text{s.t. (8) - (19)} \\
 & \omega_s * \lambda - \sum_{n=1}^{NEB} \tau_{s,n,t} * A_{n,w} + \vartheta_{s,w,t}^{(6)} - \vartheta_{s,w,t}^{(7)} = 0 \\
 & \omega_s * \mu + \tau_{s,n,t} + \vartheta_{s,n,t}^{(8)} - \vartheta_{s,n,t}^{(9)} = 0 \\
 & \omega_s * K_{i,l} + \vartheta_{s,i,t}^{(1)} - \vartheta_{s,i,t}^{(2)} - \vartheta_{s,i,t-1}^{(1)} + \vartheta_{s,i,t-1}^{(2)} - \vartheta_{s,i,t-1}^{(3)} \\
 & + \vartheta_{s,i,t-1}^{(4)} - \vartheta_{s,i,t-1,l}^{(5)} + \sum_{n=1}^{NEB} \tau_{s,n,t-1} * B_{n,i} = 0 \\
 & \text{for } t \geq 3 \\
 & \omega_s * K_{i,l} + \vartheta_{s,i,2}^{(1)} - \vartheta_{s,i,2}^{(2)} - \vartheta_{s,i,1}^{(3)} + \vartheta_{s,i,1}^{(4)} - \vartheta_{s,i,1,l}^{(5)} \\
 & + \sum_{n=1}^{NEB} \tau_{s,n,1} * B_{n,i} = 0 \\
 & \vartheta_{s,i,t}^{(1)} * \left(\sum_{l=1}^L P_{s,i,t-1,l}^g - \sum_{l=1}^L P_{s,i,t,l}^g - Ramp_i \right) = 0 \\
 & \vartheta_{s,i,t}^{(2)} * \left(\sum_{l=1}^L P_{s,i,t,l}^g - \sum_{l=1}^L P_{s,i,t-1,l}^g - Ramp_i \right) = 0 \\
 & \vartheta_{s,i,t}^{(3)} * \left(P_i^{gmin} * I_{s,i,t}^* - \sum_{l=1}^L P_{s,i,t,l}^g \right) = 0 \\
 & \vartheta_{s,i,t}^{(4)} * \left(\sum_{l=1}^L P_{s,i,t,l}^g - P_i^{gmax} * I_{s,i,t}^* \right) = 0 \\
 & \vartheta_{s,w,t}^{(6)} * (P_{s,w,t}^{wc} - P_{s,w,t}^{pre} * R_{s,w,t}^{WN}) = 0 \\
 & \vartheta_{s,n,t}^{(8)} * (P_{s,n,t}^{ec} - d_{s,n,t} * R_{s,n,t}^{EL}) = 0 \\
 & \vartheta_{s,i,t,l}^{(5)} * (-P_{s,i,t,l}^g) = 0 \quad \vartheta_{s,w,t}^{(7)} * (-P_{s,w,t}^{wc}) = 0 \\
 & \vartheta_{s,n,t}^{(9)} * (-P_{s,n,t}^{ec}) = 0.
 \end{aligned} \tag{A1}$$

where $I_{s,i,t}^*$ denotes the value of $I_{s,i,t}$ obtained from the master problem (49).

The bilinear constraints in (A1) can be transformed further through the Big-M approach:

$$\begin{cases} \vartheta_{s,i,t}^{(1)} \leq M z_{s,i,t}^{(1)} \\ \left(\text{Ramp}_i - \sum_{l=1}^L P_{s,i,t-1,l}^g + \sum_{l=1}^L P_{s,i,t,l}^g \right) \leq M \left(1 - z_{s,i,t}^{(1)} \right) \\ \vartheta_{s,i,t}^{(2)} \leq M z_{s,i,t}^{(2)} \\ \left(\text{Ramp}_i - \sum_{l=1}^L P_{s,i,t,l}^g + \sum_{l=1}^L P_{s,i,t-1,l}^g \right) \leq M \left(1 - z_{s,i,t}^{(2)} \right) \\ \vartheta_{s,i,t}^{(3)} \leq M z_{s,i,t}^{(3)} \\ \left(\sum_{l=1}^L P_{s,i,t,l}^g - P_i^{\text{gmin}} * I_{s,i,t}^* \right) \leq M \left(1 - z_{s,i,t}^{(3)} \right) \\ \vartheta_{s,i,t}^{(4)} \leq M z_{s,i,t}^{(4)} \\ \left(P_i^{\text{gmax}} * I_{s,i,t}^* - \sum_{l=1}^L P_{s,i,t,l}^g \right) \leq M \left(1 - z_{s,i,t}^{(4)} \right) \\ \vartheta_{s,w,t}^{(6)} \leq M z_{s,w,t}^{(6)} \\ (P_{s,w,t}^{\text{pre}} * R_{s,w,t}^{\text{WN}} - P_{s,w,t}^{\text{wc}}) \leq M \left(1 - z_{s,w,t}^{(6)} \right) \\ \vartheta_{s,n,t}^{(8)} \leq M z_{s,n,t}^{(8)} \\ (d_{s,n,t} * R_{s,n,t}^{\text{EL}} - P_{s,n,t}^{\text{ec}}) \leq M \left(1 - z_{s,n,t}^{(8)} \right) \\ \vartheta_{s,i,t,l}^{(5)} \leq M z_{s,i,t,l}^{(5)} P_{s,i,t,l}^g \leq M \left(1 - z_{s,i,t,l}^{(5)} \right) \\ \vartheta_{s,w,t}^{(7)} \leq M z_{s,w,t}^{(7)} P_{s,w,t}^{\text{wc}} \leq M \left(1 - z_{s,w,t}^{(7)} \right) \\ \vartheta_{s,n,t}^{(9)} \leq M z_{s,n,t}^{(9)} P_{s,n,t}^{\text{ec}} \leq M \left(1 - z_{s,n,t}^{(9)} \right) \end{cases} \quad (\text{A2})$$

Thus, the slave problem (50) becomes a mixed-integer linear programming (MILP) problem, which can be easily solved by commercially available solvers such as Gurobi.

B. Transformation of (21), (23), and (24)

The pressure drop equation (21) can be equivalently expressed as follows using the big-M method:

$$-M(1 - \gamma_{s,mn}^t) \leq q_{s,mn}^t \leq M\gamma_{s,mn}^t \quad (\text{B1})$$

$$-M(1 - \gamma_{s,mn}^t) \leq h_{s,J_m}^t - h_{s,J_n}^t - c_{mn}(q_{s,mn}^t)^2 \quad (\text{B2})$$

$$h_{s,J_m}^t - h_{s,J_n}^t - c_{mn}(q_{s,mn}^t)^2 \leq M(1 - \gamma_{s,mn}^t) \quad (\text{B3})$$

$$-M\gamma_{s,mn}^t \leq h_{s,J_m}^t - h_{s,J_n}^t + c_{mn}(q_{s,mn}^t)^2 \quad (\text{B4})$$

$$h_{s,J_m}^t - h_{s,J_n}^t + c_{mn}(q_{s,mn}^t)^2 \leq M\gamma_{s,mn}^t \quad (\text{B5})$$

M is a very large positive constant and $\gamma_{s,mn}^t$ is a binary variable. If $\gamma_{s,mn}^t = 1$, then (B1) guarantees $q_{s,mn}^t \geq 0$, (B2)–(B3) ensure that $h_{s,J_m}^t - h_{s,J_n}^t - c_{mn}(q_{s,mn}^t)^2 = 0$, and (B4)–(B5) hold. If $\gamma_{s,mn}^t = 0$, the flow changes direction ($q_{s,mn}^t \leq 0$), (B4)–(B5) ensure that $h_{s,J_m}^t - h_{s,J_n}^t + c_{mn}(q_{s,mn}^t)^2 = 0$ and (B2)–(B3) hold.

The same process can also be applied to the “on/off” statement in constraints (23) and (24):

$$-M(1 - r_{s,P_m}^t) \leq \hat{h}_{s,P_m}^t - (a_{P_m}(q_{s,P_m}^t)^2 + b_{P_m}q_{s,P_m}^t + c_{P_m}) \leq M r_{s,P_m}^t \quad (\text{B6})$$

$$\hat{h}_{s,P_m}^t - (a_{P_m}(q_{s,P_m}^t)^2 + b_{P_m}q_{s,P_m}^t + c_{P_m}) \leq M(1 - r_{s,P_m}^t) \quad (\text{B7})$$

$$q_{P_m}^{\text{min}} r_{s,P_m}^t \leq q_{s,P_m}^t \leq q_{P_m}^{\text{max}} r_{s,P_m}^t \quad (\text{B8})$$

r_{s,P_m}^t is a binary variable. If $r_{s,P_m}^t = 1$, then the pump is on, (B6)–(B7) ensure $\hat{h}_{s,P_m}^t - (a_{P_m}(q_{s,P_m}^t)^2 + b_{P_m}q_{s,P_m}^t + c_{P_m}) = 0$ and (B8) holds. If $r_{s,P_m}^t = 0$, then the pump is off, (B8) ensures that $q_{s,P_m}^t = 0$ and (B6)–(B7) hold.

C. Diagram of the IEEE-RTS 24 Power System

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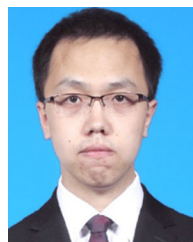
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